

India CPI Inflation Case Study

Overview:-

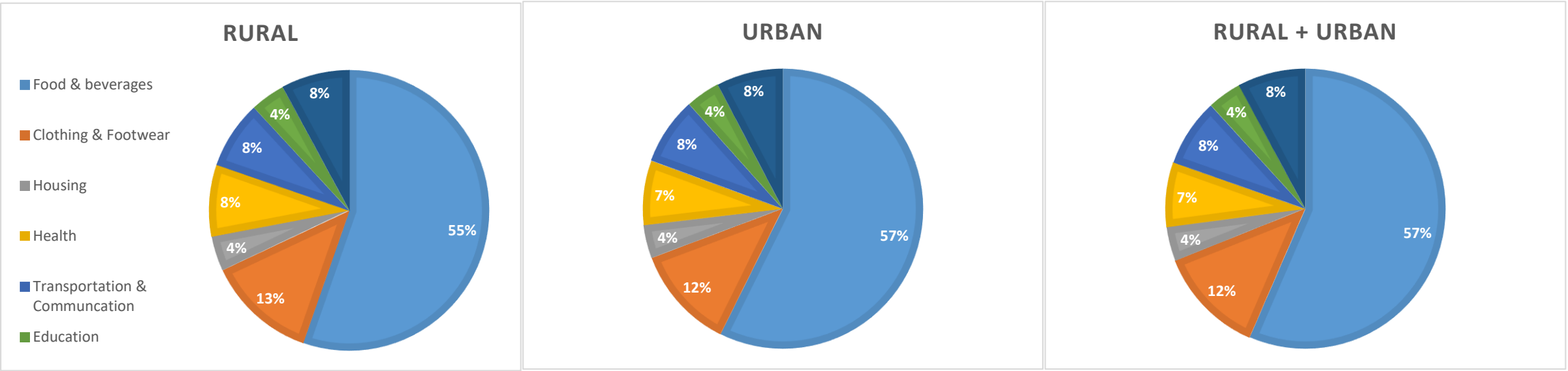
- In India, the Consumer Price Index (CPI) is used to measure inflation, and it involves a fixed basket of goods and services. This basket is comprehensive and includes a wide array of items that an average Indian consumer uses. These items are not limited to just food and clothing but extend to transportation, medical care, electricity, education, and almost every other category that involves expenditure of money.
- The CPI is calculated by comparing the general price level in the markets during a particular time period with a base year. The items in the CPI basket are classified across various categories like food and beverages, clothing, housing, fuel and light, and recreation, among others.
- The CPI basket contains categories like food and beverages, housing, apparel, transportation, medical care, and more. The weight of each category in the total index might differ based on its relative importance to the average consumer expenditure but for the purpose of this analysis consider equal weights across all categories.
- The dataset provided is a CPI inflation index extracted from GOI website
- Each number represents the index value for that month and category

Sub-category	Category
Cereals and products	Food & Beverages
Meat and fish	
Egg	
Milk and products	
Oils and fats	
Fruits	
Vegetables	
Pulses and products	
Sugar and Confectionery	
Spices	
Non-alcoholic beverages	
Prepared meals, snacks, sweets etc.	
Food and beverages	
Pan, tobacco and intoxicants	
Clothing	Clothing & Footwear
Footwear	
Clothing and footwear	
Housing	Housing
Household goods and services	
Health	Health
Personal care and effects	
Transportation & Comm	Transportation &
Fuel and light	
Education	Education
Recreation and amusement	Recreation & Misc
Miscellaneous	

Contribution of Broader Categories towards CPI Basket

\\To get the highest contributor for last month we need to add all index values as specified in the category sheet for latest month

Categories	Rural	Contribution	Urban	Contribution	Rural+Urban	Contribution
Food & beverages	2490.6	55.36	2539.3	57.35	2507.9	56.51
Clothing & Footwear	569.9	12.67	528.7	11.94	553.2	12.47
Housing	179.8	4.00	170.1	3.84	175.2	3.95
Health	372.7	8.28	330.4	7.46	330	7.44
Transportation & Communcation	352.2	7.83	343.8	7.76	347.6	7.83
Education	180.3	4.01	174.8	3.95	177.1	3.99
Recreation & Misc	353.3	7.85	340.8	7.70	346.9	7.82
Total	4498.8	100.00	4427.9	100.00	4437.9	100.00



Food & beverage is the highest contribution towards CPI calculation in all 3 sectors
Clothing & Footwear and housing are the mid contributors to CPI calculation
whereas other category do contribute but not that much.

Trend of Y-o-Y increase in CPI (Rural + Urban) inflation starting from 2017 for the entire basket of products combined

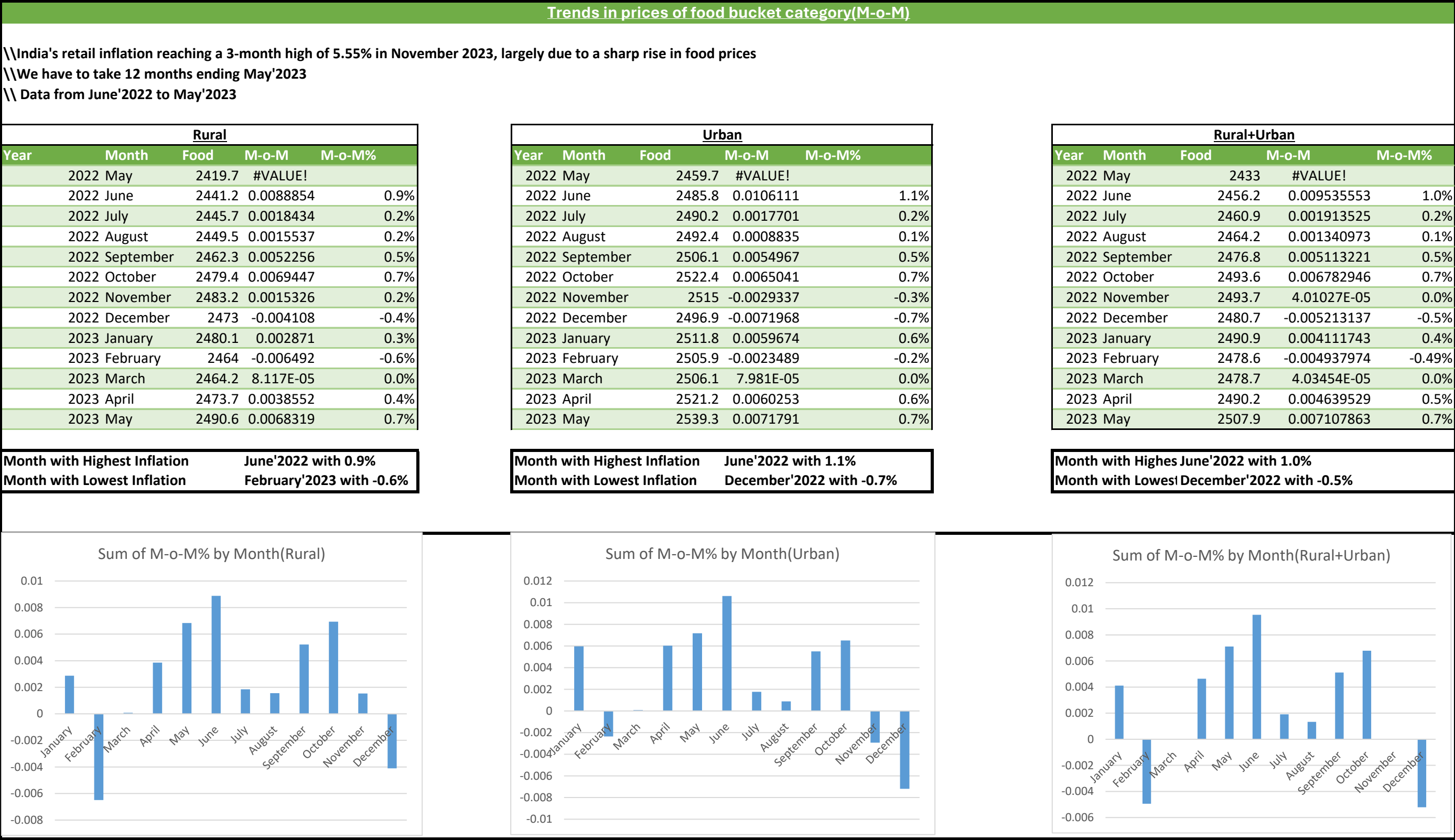
\\We have data from 2013 Jan to 2023 May
\\So we will take 2017 May to 2023 May
\\We will consider 2017 May also to get June data

Sector	Year	Month	General index	Y-o-Y Calculation	Y-o-Y%
Rural+Urban	2016	May	128.6		
Rural+Urban	2017	May	131.4	0.021772939	2.2%
Rural+Urban	2018	May	137.8	0.04870624	4.9%
Rural+Urban	2019	May	142	0.030478955	3.0%
Rural+Urban	2020	May	151.1	0.064377934	6.4%
Rural+Urban	2021	May	160.4	0.061255996	6.1%
Rural+Urban	2022	May	171.7	0.070448878	7.0%
Rural+Urban	2023	May	179.1	0.043098427	4.3%

For year 2022, inflation rate is highest at 7%

As per research, due to Russia-Ukraine wars which disrupted the supply chains and because of that inflation rate is highest in 2022.

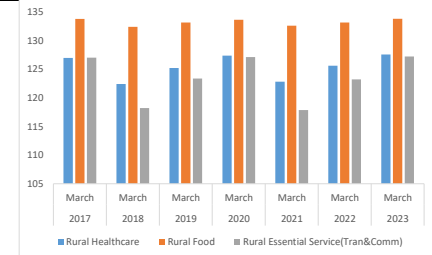




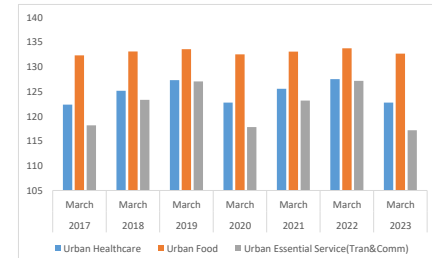
COVID-19 effects on India CPI

\\As we have enough data, we will taking 3yrs back 2019 and 3yrs after 2019
 \\Consider Mar'20 as the onset of covid, and can compare the inflation trend before and after Mar'20 to see if there is a change in inflation % before and after
 \\We will be considering Healthcare,Food and Essential Service

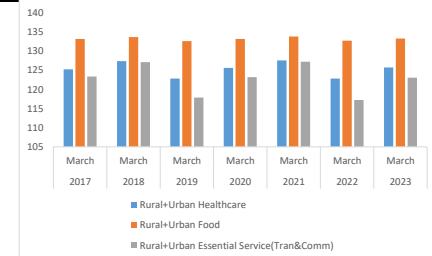
Rural						
Year	Month	Healthcare	Food	Essential Service(Tran&Comm)	Metric	Value
2017	March	126.95		133.76	Healthcare Avg (Pre-COVID 2017-2019)	124.85
2018	March	122.4		132.37	Healthcare Avg (Post-COVID 2020-2023)	125.825
2019	March	125.2		133.14	Food Avg (Pre-COVID 2017-2019)	133.09
2020	March	127.35		133.62	Food Avg (Post-COVID 2020-2023)	133.28
2021	March	122.8		132.58	Essential Svc Avg (Pre-COVID)	122.85
2022	March	125.6		133.13	Essential Svc Avg (Post-COVID)	123.8375
2023	March	127.55		133.79	Healthcare Change %	0.781
Healthcare- Only a marginal increase of 0.78% post-COVID, indicating stable healthcare cost indices in rural areas					Food Change %	0.140
Food- Minimal increase of 0.14%, suggesting effective supply chain management despite pandemic disruptions					Essential Svc Change %	0.804
Essential Service- 0.80% increase, likely due to increased fuel and transportation costs					Peak Healthcare Year	2023
					Peak Food Year	2023
					Peak Essential Svc Year	2023



Urban						
Year	Month	Healthcare	Food	Essential Service(Tran&Comm)	Metric	Value
2017	March	122.4		132.37	Healthcare Avg (Pre-COVID 2017-2019)	124.9833333
2018	March	125.2		133.14	Healthcare Avg (Post-COVID 2020-2023)	124.6875
2019	March	127.35		133.62	Food Avg (Pre-COVID 2017-2019)	133.05
2020	March	122.8		132.58	Food Avg (Post-COVID 2020-2023)	133.05
2021	March	125.6		133.13	Essential Svc Avg (Pre-COVID)	122.8833333
2022	March	127.55		133.79	Essential Svc Avg (Post-COVID)	121.3625
2023	March	122.8		132.71	Healthcare Change %	-0.237
HealthCare- Unusual -0.24% decline post-COVID, possibly due to increased digital health adoption and competition					Food Change %	0.004
Food- Near-zero change (0.004%), showing robust urban supply chains					Essential Svc Change %	-1.238
Essential Service- Dropped significantly -1.24% decrease is notable, likely due to reduced travel demand during lockdowns and WFH culture					Peak Healthcare Year	2022
					Peak Food Year	2022
					Peak Essential Svc Year	2022



Rural+Urban						
Year	Month	Healthcare	Food	Essential Service(Tran&Comm)	Metric	Value
2017	March	125.2		133.14	Healthcare Avg (Pre-COVID 2017-2019)	125.1166667
2018	March	127.35		133.62	Healthcare Avg (Post-COVID 2020-2023)	125.4125
2019	March	122.8		132.58	Food Avg (Pre-COVID 2017-2019)	133.11
2020	March	125.6		133.13	Food Avg (Post-COVID 2020-2023)	133.22
2021	March	127.55		133.79	Essential Svc Avg (Pre-COVID)	122.7666667
2022	March	122.8		132.71	Essential Svc Avg (Post-COVID)	122.6625
2023	March	125.7		133.27	Healthcare Change %	0.236445984
HealthCare- Modest 0.24% rise reflects balanced national healthcare cost management					Food Change %	0.081830865
Food- Modest 0.24% rise reflects balanced national healthcare cost management					Essential Svc Change %	-0.084849308
Essential Service- Marginal -0.08% decrease balances rural increases with urban decreases					Peak Healthcare Year	2021
					Peak Food Year	2021
					Peak Essential Svc Year	2021



Trends in oil price change with change in inflation prices of all the categories																												
\\Imported Crude oil data from PPAC site																												
2021 May 2023 May																												
		Rural									Urban									Rural+Urban								
		Categories									Categories									Categories								
M.Y	Imported Crude Oil price	Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index	Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index	Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index			
Apr-21	63	2236.3	475.7	294.62	319.9	151.7	164.1	316.6	157.6	2284	448.6	309	314.1	142.1	157.6	299.6	158	2252.9	464.6	313.2	317.7	146.6	160.3	307.6	157.8			
May-21	67	2284.9	490.4	297.92	328.4	153.2	167.6	321.1	161.1	2322.9	450.8	311.7	317.9	145	156.6	304.9	159.5	2297.6	474.3	316.3	324.4	148.9	161.2	312.6	160.4			
Jun-21	72	2311.7	489.8	298.32	329.1	154.2	166.8	321.9	162.1	2349.8	452.6	310.3	318.8	147.5	158.1	304.1	160.4	2324.7	474.7	315.3	325.1	150.7	161.7	312.5	161.3			
Jul-21	74	2322.1	492.4	299.42	330.8	157.1	167.2	323.5	163.2	2367.3	455.3	312.2	321.1	149.5	160.3	306.2	161.8	2338.2	477.3	317.3	327.1	153.1	163.2	314.3	162.5			
Aug-21	70	2321	495.9	300.02	331.4	157.7	167.5	324.4	163.6	2354.4	460.7	315.3	322.4	150.4	160.4	309.7	162.3	2334.1	483	319.6	328.4	154	163.8	317.6	163.2			
Sep-21	73	2324.1	498.4	300.42	332.1	157.8	168.5	326.5	164	2354.4	460.8	315.4	322.4	150.5	160.3	309.9	162.3	2334.1	483.2	319.6	328.4	154	163.7	317.7	163.2			
Oct-21	82	2355.4	502	301.12	333.6	159.5	169	327.9	166.3	2395.4	463.5	317.9	323.8	152.2	160.3	312.1	164.6	2368.2	486.3	322	329.9	155.7	163.9	319.6	165.5			
Nov-21	81	2373.4	506.2	302.02	335.8	158.9	169.3	329	167.6	2414.9	467.3	319.4	326	151.2	160.8	314	165.6	2387	490.4	323.5	332.1	154.8	164.3	321.2	166.7			
Dec-21	73	2359	510.3	303.02	336.8	160.1	169.7	330.5	167	2403.1	470.7	319.4	327.5	151.8	160.6	315.4	165.2	2373.3	494.2	323.6	333.2	155.7	164.4	322.6	166.2			
Jan-22	85	2343.7	515.2	304.02	337.9	160.8	169.9	331.5	166.4	2383.1	475.4	321.3	328.9	152.7	161	317	165	2356.4	499.1	325.6	334.4	156.5	164.7	323.9	165.7			
Feb-22	94	2341.9	518.8	304.82	339.8	161.2	170.3	332.8	166.7	2380	479.5	322.9	331.4	153.1	162	318.9	165.5	2354	502.8	327.3	336.6	156.9	165.4	325.6	166.1			
Mar-22	113	2371.4	523.7	305.62	343.4	162	170.6	334.9	168.7	2393.8	484.6	323.9	335	154.2	162.7	321.4	166.5	2377.9	507.8	328.1	340.2	157.9	166	327.9	167.7			
Apr-22	103	2399.4	529.7	306.82	346	166.2	170.9	337.4	170.8	2427.5	489.2	326.8	337.4	159.3	164	325.3	169.2	2408.2	513.2	331	342.8	162.6	166.9	331.2	170.1			
May-22	110	2419.7	535.5	308.02	346.2	167.1	171.8	338.5	172.5	2459.7	493.7	328.6	338.3	159.4	165.2	327	170.8	2433	518.6	332.7	343.2	163	167.9	332.6	171.7			
Jun-22	116	2441.2	539.8	309.42	347.7	165.5	172.6	339	173.6	2485.8	498.4	328.9	340.1	157.2	166.5	327.9	171.4	2456.2	523	333.2	344.8	161.1	169	333.3	172.6			
Jul-22	105	2445.7	544	310.42	348.5	166.3	174.7	340.4	174.3	2490.2	502	330.9	341.5	157.4	169.1	329.3	172.3	2460.9	526.9	335.2	345.8	161.6	171.4	334.7	173.4			
Aug-22	97	2449.5	547.9	311.42	350.5	166.6	175.7	341.9	175.3	2492.4	505.3	333.2	344	157.7	169.9	330.5	173.1	2464.2	530.7	337.5	348	161.9	172.3	336	174.3			
Sep-22	91	2462.3	552.5	312.72	351	166.9	176.2	343.1	176.4	2506.1	509.7	334.5	344.9	158.2	170.9	331.9	174.1	2476.8	535.1	339	348.7	162.3	173.1	337.3	175.3			
Oct-22	92	2479.4	556.4	313.52	353.2	167.4	176.5	344.5	177.9	2522.4	511.7	337.2	347	158.8	171.2	333.1	175.3	2493.6	538.2	341.6	350.8	162.9	173.4	338.7	176.7			
Nov-22	88	2483.2	559.3	314.62	355.7	167.5	176.9	345.4	177.8	2515	514.9	338.7	349.6	158.9	171.5	334.1	174.1	2493.7	541.4	343.2	353.4	163	173.7	339.6	176.5			
Dec-22	78	2473	561.8	315.52	359.2	167.8	177.3	346.7	177.1	2496.9	517.9	338	353.2	159.4	171.8	335.3	174.1	2480.7	544	342.8	356.9	163.4	174.1	340.9	175.7			
Jan-23	81	2480.1	563.9	316.32	363.1	168.2	177.8	348.3	177.8	2511.8	520.6	340.1	357.3	159.5	171.8	336.7	174.9	2490.9	546.3	345	360.9	163.6	174.3	342.3	176.5			
Feb-23	82	2464	566.6	317.72	367.3	169	178.5	350.7	178	2505.9	525.5	342.7	362.2	159.8	172.5	338.4	176.3	2478.6	550	347.7	365.4	164.2	175	344.4	177.2			
Mar-23	79	2464.2	566.6	317.72	367.3	169	178.5	350.7	178	2506.1	525.4	342.7	362.3	159.8	172.5	338.4	176.3	2478.7	549.9	347.7	365.4	164.2	175	344.4	177.2			
Apr-23	84	2473.7	568.2	318.22	371	169.4	179.4	352.1	178.8	2521.2	527.6	344.8	365.9	160.1	174.2	339.7	177.4	2490.2	551.8	349.8	369	164.5	176.4	345.7	178.1			
May-23	75	2490.6	569.9	318.92	372.7	169.7	180.3	353.3	179.8	2539.3	528.7	345.7	367.8	160.4	174.8	340.8	178.2	2507.9	553.2	350.8	370.9	164.8	177.1	346.9	179.1			
\\Correlating each category with Crude oil from different sectors to know the affect.																												
		Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index	Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index	Food & beverages	Clothing & Footwear	Housing	Health	Transportation & Communcation	Education	Recreation & Misc	General Index			
Correl		0.4204	0.3746	0.3066	0.2538	0.4965	0.2474	0.3514	0.3931	0.4026	0.3502	0.3080	0.2424	0.5358	0.2733	0.4103	0.3808	0.4111	0.3640	0.2925	0.2484	0.5156	0.2603	0.3846	0.3868			
Rural									Urban									Rural+Urban										
In all sectors, Transportation & Comm had the highest affect with change in crude oil prices																												